Prompt:

I'll give you some process rules in the form of sentences and I want you to: 1- extract process activities and have letter labeling for each, the labelling is either A or B. I'll give you the list of activities that you need to replace with either A or B. 2- create LTL constraints for each sentence. The LTL constrains falls into 5 categories:

1- initialize constraints: Init(A) where A is the first to activity occur (Start -> A). An example is "Every claim has to be created in the beginning." Where A is create claim.

2- end constraints: End(A) where A is the last activity to occur (End -> A). An example is "After a claim has been paid, it is closed." Where A is pay claim.

3- response constraints: Response (A, B) If A occurs, then B occurs after A (A -> F(B)). An example is "Once a claim has been approved, it should eventually be paid out." Where A is approve claim and B is pay out claim.

4- precedence constraints: Precedence (A, B) B occurs only if preceded by A (NOT(b) U A). An example is "A claim should be created before it can be approved." Where A is create claim and B is approve claim

5- succession constraints: Succession (A, B) A occurs if and only if B occurs after A, (A -> F(B)) AND (NOT(B) U A). An example is "Once a claim has been rejected, it should not be paid out." Where A is reject claim and B is not pay out claim

Do this sentence by sentence, tell me what the activities and the LTL constraints of the sentence are. If the sentence has multiple constraints, please provide them.

Response:

Sure, I'd be happy to help you with that. However, in order to proceed, I need the sentences you'd like me to work with, as well as the list of activities that you want to replace with "A" or "B". Once I have this information, I'll be able to extract the process activities, label them, and create the corresponding LTL constraints.

Prompt:

here's the list of activities that you need to replace with either A or B:

create claim

creation claim

approve claim

reject claim

payout

pay out claim

create case

arrange meeting

enter metadata

submit case

assign case id

agree date

examine patient

check x-ray risk

perform x-ray

perform surgery

apply cast

book ticket

payment ticket

check-in flight

receive invoice

decide invoice legit

receive all necessary approvals

confirm payment

receipt of an invoice

enter data

pay invoice

send request

give confirmation

register client data

provide room service

provide laundry service

leave

search report

bring in a defective computer

each of these activities

error is detected

register claim

Check recommendation

Mark as not ok

Mark as Ok

Receive request for new hardware

Analyze request

Deny request

Approve request

Receive offer

this is accepted

take care of invoice

receive customer request

need further review

submit office supply request

register requirement

reject request

ask to make change

select vendor

approval, rejection, change

submit vacation request

review order

inform customer

wait for x days

inform about delay

deliver package

Receive order

reserve part

ship bicycle to the customer

Receive amount request

Request approval

File expense

Receive expense report

check take place

supervisor approves

supervisor rejects

respond

approval claim

pay claim

case closed manually

manage case

hold meeting

inform

get local id

propose dates

perform x-ray risk

perform reposition

prescribe rehabilitation

remove cast

provide personal information

booking phase completion

money transfer

modify provided data

cancellation

enter all relevant data

close case

give approval

prepare proposal

proposal

bill

charge bill

put in a file together with the claim documentation

analyze the defect

test proper system functionality

execute repair activity

examine claim

send it back

Proceed with claim

inform employee via email

edit request

fill in form

decide if approved

receive hardware by IT department

Send back request to IT

receive request

return to petitioner/employee

go to purchase department

generate purchase order

send notification

return application

generate notification

order missing products

check again

process delivery

update system

assemble bicycle

Assess credit risk

Accept credit

Perform check

stop process

deposit of reimbursement

receives a notification

send email

Response:
Great, now we need the sentences you want me to process using these activities. Could you please provide the sentences for analysis?

Prompt:

0	A claim should be created, before it can be approved.
1	A claim must be created, before it is approved.
2	When a claim is created, it may be approved.
3	If a claim is created, it can be approved.
4	They have to create the claim, before they can approve it.
5	Only after a claim is created, it is possible to approve the claim.
6	If a claim is approved, then it must have been created first.
7	To approve a claim, it should be created first.
8	A claim can be approved, after it has been created.
9	Approving the claim can happen at anytime, unless it still needs to be created.
10	Creation of the claim is required, before it can be approved.

ChatGPT

Thank you for providing the sentences. Let's label the activities and create LTL constraints.

"A claim should be created, before it can be approved."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"A claim must be created, before it is approved."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"When a claim is created, it may be approved."

Activities:

create claim: A

approve claim: B

LTL constraint: Response (A, B)

"If a claim is created, it can be approved."

Activities:

create claim: A

approve claim: B

LTL constraint: Response (A, B)

"They have to create the claim, before they can approve it."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"Only after a claim is created, it is possible to approve the claim."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"If a claim is approved, then it must have been created first."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"To approve a claim, it should be created first."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"A claim can be approved, after it has been created."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"Approving the claim can happen at anytime, unless it still needs to be created."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"Creation of the claim is required, before it can be approved."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

Please note, all the sentences you provided have the activities "create claim" and "approve claim" which are associated with the LTL constraint of Precedence (A, B). There can be a possibility of multiple constraints in case of multiple activities or varied conditions. But in the given sentences, there are only two activities in a defined order, so they fall under the Precedence category.

Prompt:

do the same for the following:

11 The creation of the claim is a condition for the approval of the claim.

12 Only after the creation of a claim, the approval of the claim is possible.

13 When approval of the claim occurs, creation of the claim must have preceded.

14 If approval of the claim occurs, then creation of the claim must have been carried out first.

15 The approval of the claim requires that the creation of the claim occurs before it.

16 The approval of the claim should be preceded by the creation of the claim.

17 A claim must be created, before it can be approved or rejected.

18 A claim should be approved before it can be paid out.

19 Once a claim has been approved, it should eventually be paid out.

20 Once a claim has been rejected, it should not be paid out.

Response:

Sure, let's proceed with labeling activities and creating LTL constraints for the next set of sentences.

"The creation of the claim is a condition for the approval of the claim."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"Only after the creation of a claim, the approval of the claim is possible."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"When approval of the claim occurs, creation of the claim must have preceded."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"If approval of the claim occurs, then creation of the claim must have been carried out first."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"The approval of the claim requires that the creation of the claim occurs before it."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"The approval of the claim should be preceded by the creation of the claim."

Activities:

create claim: A

approve claim: B

LTL constraint: Precedence (A, B)

"A claim must be created, before it can be approved or rejected."

Activities:

create claim: A

approve claim: B

reject claim: C

LTL constraints: Precedence (A, B) and Precedence (A, C)

"A claim should be approved before it can be paid out."

Activities:

approve claim: A

pay out claim: B

LTL constraint: Precedence (A, B)

"Once a claim has been approved, it should eventually be paid out."

Activities:

approve claim: A

pay out claim: B

LTL constraint: Response (A, B)

"Once a claim has been rejected, it should not be paid out."

Activities:

reject claim: A

not pay out claim: B

LTL constraint: Succession (A, B)

These sentences have the activities "create claim", "approve claim", "reject claim", and "pay out claim", which are associated with the LTL constraints of Precedence, Response, and Succession depending on the specific order and conditions mentioned in each sentence.

Prompt:
do the same for the following:

21 Payout should end the process and therefore exclude everything.

22 A case can be closed manually at any time, unless we still need to pay out the claim.

23 The case can be managed at the union after it has been created.

24 After a case is created, management can and must arrange a meeting between the union case worker and the other case worker.

25 After a meeting is arranged it must be held.

26 The case worker should enter metadata on the case, inform about when he is available for participating in a meeting and then submit the case.

27 When a case is submitted it may get a local id at the union, but it should also subsequently be assigned a case id in LO.

28 When a case is submitted, LO should eventually propose dates.

29 Only after LO has assigned its case id it is possible to manage the case and for LO to propose dates.

30 Once a date has been agreed upon a meeting should eventually be held.

Response:
Sure, let's proceed with labeling activities and creating LTL constraints for the next set of sentences.

"Payout should end the process and therefore exclude everything."

Activities:

pay out claim: A

LTL constraint: End (A)

"A case can be closed manually at any time, unless we still need to pay out the claim."

Activities:

pay out claim: A

case closed manually: B

LTL constraint: Precedence (A, B)

"The case can be managed at the union after it has been created."

Activities:

create case: A

manage case: B

LTL constraint: Precedence (A, B)

"After a case is created, management can and must arrange a meeting between the union case worker and the other case worker."

Activities:

create case: A

arrange meeting: B

LTL constraint: Response (A, B)

"After a meeting is arranged it must be held."

Activities:

arrange meeting: A

hold meeting: B

LTL constraint: Response (A, B)

"The case worker should enter metadata on the case, inform about when he is available for participating in a meeting and then submit the case."

Activities:

enter metadata: A

inform: B

submit case: C

LTL constraints: Precedence (A, B), Precedence (B, C)

"When a case is submitted it may get a local id at the union, but it should also subsequently be assigned a case id in LO."

Activities:

submit case: A

get local id: B

assign case id: C

LTL constraints: Response (A, B), Response (A, C)

"When a case is submitted, LO should eventually propose dates."

Activities:

submit case: A

propose dates: B

LTL constraint: Response (A, B)

"Only after LO has assigned its case id it is possible to manage the case and for LO to propose dates."

Activities:

assign case id: A

manage case: B

propose dates: C

LTL constraints: Precedence (A, B), Precedence (A, C)

"Once a date has been agreed upon a meeting should eventually be held."

Activities:

agree date: A

hold meeting: B

LTL constraint: Response (A, B)

These sentences have various activities like "pay out claim", "case closed manually", "create case", "manage case", "arrange meeting", "hold meeting", "enter metadata", "inform", "submit case", "get local id", "assign case id", "propose dates", and "agree date", which are associated with the LTL constraints of End, Precedence, Response depending on the specific order and conditions mentioned in each sentence.

Prompt:
do the same for the following:

31 Every process instance starts by examining a patient.

32 If an x-ray is taken, then the x-ray risk must be checked before it

33 Performing a reposition, applying a cast and performing surgery require that x-rays are performed before.

34 If a surgery is performed, then prescription of rehabilitation occurs eventually after it.

35 After a cast is applied, eventually the cast is removed and vice versa, before every removal of a cast, a cast must be applied.

36 The process begins with the booking of the ticket

37 Personal information of passengers can be provided after the booking of the ticket

38 The payment of the ticket triggers the completion of the booking phase.

39 The payment of the ticket is eventually followed by the actual transfer of money

40 As long as the Check-in of the flight does not take place, the customers can still modify the provided data to change

Response:
Sure, let's proceed with labeling activities and creating LTL constraints for the next set of sentences.

"Every process instance starts by examining a patient."

Activities:

examine patient: A

LTL constraint: Init (A)

"If an x-ray is taken, then the x-ray risk must be checked before it."

Activities:

check x-ray risk: A

perform x-ray: B

LTL constraint: Precedence (A, B)

"Performing a reposition, applying a cast and performing surgery require that x-rays are performed before."

Activities:

perform x-ray: A

perform reposition: B

apply cast: C

perform surgery: D

LTL constraints: Precedence (A, B), Precedence (A, C), Precedence (A, D)

"If a surgery is performed, then prescription of rehabilitation occurs eventually after it."

Activities:

perform surgery: A

prescribe rehabilitation: B

LTL constraint: Response (A, B)

"After a cast is applied, eventually the cast is removed and vice versa, before every removal of a cast, a cast must be applied."

Activities:

apply cast: A

remove cast: B

LTL constraints: Response (A, B), Precedence (A, B)

"The process begins with the booking of the ticket."

Activities:

book ticket: A

LTL constraint: Init (A)

"Personal information of passengers can be provided after the booking of the ticket."

Activities:

book ticket: A

provide personal information: B

LTL constraint: Precedence (A, B)

"The payment of the ticket triggers the completion of the booking phase."

Activities:

payment ticket: A

booking phase completion: B

LTL constraint: Response (A, B)

"The payment of the ticket is eventually followed by the actual transfer of money."

Activities:

payment ticket: A

money transfer: B

LTL constraint: Response (A, B)

"As long as the Check-in of the flight does not take place, the customers can still modify the provided data to change."

Activities:

check-in flight: A

modify provided data: B

LTL constraint: Precedence (B, A)

These sentences have various activities like "examine patient", "check x-ray risk", "perform x-ray", "perform reposition", "apply cast", "perform surgery", "prescribe rehabilitation", "remove cast", "book ticket", "provide personal information", "payment ticket", "booking phase completion", "money transfer", "check-in flight", and "modify provided data", which are associated with the LTL constraints of Init, Precedence, Response depending on the specific order and conditions mentioned in each sentence.

Prompt:

do the same for the following:

41 After check-in of the flight, only the cancellation is admitted

42 The process starts when an invoice is received by the administration department.

43 If they decide that the invoice appears legit then they enter all relevant data into the system

44 When all necessary approvals have been received the invoice can be paid.

45 Once payment is confirmed, the invoice case should be closed.

46 The receipt of an invoice is required before the administration department can enter data or scan the invoice.

47 Enter data is required before any approval can be given.

48 Finally, we have to pay the invoice before we can confirm payment.

49 If a request is sent, then a proposal is expected to be prepared afterwards

50 A confirmation is supposed to be mandatorily given after the proposal, and vice-versa any proposal is expected to precede a confirmation

Response:
Sure, let's proceed with labeling activities and creating LTL constraints for the next set of sentences.

"After check-in of the flight, only the cancellation is admitted."

Activities:

check-in flight: A

cancellation: B

LTL constraint: Response (A, B)

"The process starts when an invoice is received by the administration department."

Activities:

receive invoice: A

LTL constraint: Init (A)

"If they decide that the invoice appears legit then they enter all relevant data into the system."

Activities:

decide invoice legit: A

enter all relevant data: B

LTL constraint: Response (A, B)

"When all necessary approvals have been received the invoice can be paid."

Activities:

receive all necessary approvals: A

pay invoice: B

LTL constraint: Response (A, B)

"Once payment is confirmed, the invoice case should be closed."

Activities:

confirm payment: A

close case: B

LTL constraint: Response (A, B)

"The receipt of an invoice is required before the administration department can enter data or scan the invoice."

Activities:

receipt of an invoice: A

enter data: B

LTL constraint: Precedence (A, B)

"Enter data is required before any approval can be given."

Activities:

enter data: A

give approval: B

LTL constraint: Precedence (A, B)

"Finally, we have to pay the invoice before we can confirm payment."

Activities:

pay invoice: A

confirm payment: B

LTL constraint: Precedence (A, B)

"If a request is sent, then a proposal is expected to be prepared afterwards."

Activities:

send request: A

prepare proposal: B

LTL constraint: Response (A, B)

"A confirmation is supposed to be mandatorily given after the proposal, and vice-versa any proposal is expected to precede a confirmation."

Activities:

prepare proposal: A

give confirmation: B

LTL constraint: Succession (A, B)

These sentences have various activities like "check-in flight", "cancellation", "receive invoice", "decide invoice legit", "enter all relevant data", "receive all necessary approvals", "pay invoice", "confirm payment", "close case", "receipt of an invoice", "enter data", "give approval", "send request", "prepare proposal", and "give confirmation", which are associated with the LTL constraints of Init, Response, Precedence, and Succession depending on the specific order and conditions mentioned in each sentence.

Prompt:
do the same for the following:

51 Every process instance has to start by registering client data

52 Every provided room service must be billed.

53 Every provided laundry service must be billed

54 When the client leaves, the bill must be charged

55 The Police Report related to the car accident is searched within the Police Report database and put in a file together with the Claim Documentation.

56 A customer brings in a defective computer and the manager analyzes the defect and gives back an invoice.

57 After each of these activities, the proper system functionality is tested.

58 If an error is detected, another arbitrary repair activity is executed, otherwise the repair is finished.

59 After a claim is registered, it is examined by a claims officer.

60 This recommendation is then checked by a senior claims officer who may mark the claim as OK or Not OK.

Response:
Sure, let's label these activities and form the LTL constraints.

"Every process instance has to start by registering client data."

Activities:

register client data: A

LTL constraint: Init (A)

"Every provided room service must be billed."

Activities:

provide room service: A

bill: B

LTL constraint: Response (A, B)

"Every provided laundry service must be billed."

Activities:

provide laundry service: A

bill: B

LTL constraint: Response (A, B)

"When the client leaves, the bill must be charged."

Activities:

leave: A

charge bill: B

LTL constraint: Response (A, B)

"The Police Report related to the car accident is searched within the Police Report database and put in a file together with the Claim Documentation."

Activities:

search report: A

put in a file together with the claim documentation: B

LTL constraint: Response (A, B)

"A customer brings in a defective computer and the manager analyzes the defect and gives back an invoice."

Activities:

bring in a defective computer: A

analyze the defect: B

give back invoice: C

LTL constraint: Response (A, B), Response (B, C)

"After each of these activities, the proper system functionality is tested."

Activities:

each of these activities: A

test proper system functionality: B

LTL constraint: Response (A, B)

"If an error is detected, another arbitrary repair activity is executed, otherwise the repair is finished."

Activities:

error is detected: A

execute repair activity: B

LTL constraint: Response (A, B)

"After a claim is registered, it is examined by a claims officer."

Activities:

register claim: A

examine claim: B

LTL constraint: Response (A, B)

"This recommendation is then checked by a senior claims officer who may mark the claim as OK or Not OK."

Activities:

Check recommendation: A

Mark as Ok: B

Mark as not Ok: C

LTL constraint: Response (A, B or C)

These sentences have activities like "register client data", "provide room service", "bill", "provide laundry service", "leave", "charge bill", "search report", "put in a file together with the claim documentation", "bring in a defective computer", "analyze the defect", "give back invoice", "each of these activities", "test proper system functionality", "error is detected", "execute repair activity", "register claim", "examine claim", "Check recommendation", "Mark as Ok", "Mark as not Ok", which are associated with the LTL constraints of Init and Response depending on the specific order and conditions mentioned in each sentence.

Prompt:
do the same for the following:

61 If the claim is marked as Not OK, it is sent back to the claims officer and the recommendation is repeated.

62 If the claim is OK, the claim handling process proceeds.

63 The process of buying new hardware is started as soon as a request for new hardware is received by the IT department.

64 The next step is for the IT department to analyse the request and either approve or deny it.

65 If it is denied, the employee is informed via email about the denial and the process is ended.

66 However, if the request is approved it is edited and completed.

67 Once this offer is received, a form is filled in and sent to the management.

68 The management then analyses the request and decides if it is approved or not.

69 When this is accepted and the financial resources are found, the hardware is received by the IT department.

70 After which the financial department will take care of the invoice and the processes is ended.

Response:
Sure, let's label these activities and form the LTL constraints.

"If the claim is marked as Not OK, it is sent back to the claims officer and the recommendation is repeated."

Activities:

Mark as not Ok: A

send it back: B

Check recommendation: C

LTL constraint: Response (A, B), Response (B, C)

"If the claim is OK, the claim handling process proceeds."

Activities:

Mark as Ok: A

Proceed with claim: B

LTL constraint: Response (A, B)

"The process of buying new hardware is started as soon as a request for new hardware is received by the IT department."

Activities:

Receive request for new hardware: A

LTL constraint: Init (A)

"The next step is for the IT department to analyse the request and either approve or deny it."

Activities:

Analyze request: A

Approve request: B

Deny request: C

LTL constraint: Response (A, B or C)

"If it is denied, the employee is informed via email about the denial and the process is ended."

Activities:

Deny request: A

inform employee via email: B

LTL constraint: Response (A, B)

"However, if the request is approved it is edited and completed."

Activities:

Approve request: A

edit request: B

LTL constraint: Response (A, B)

"Once this offer is received, a form is filled in and sent to the management."

Activities:

Receive offer: A

fill in form: B

LTL constraint: Response (A, B)

"The management then analyses the request and decides if it is approved or not."

Activities:

Analyze request: A

decide if approved: B

LTL constraint: Response (A, B)

"When this is accepted and the financial resources are found, the hardware is received by the IT department."

Activities:

this is accepted: A

receive hardware by IT department: B

LTL constraint: Response (A, B)

"After which the financial department will take care of the invoice and the processes is ended."

Activities:

take care of invoice: A

LTL constraint: End (A)

These sentences have activities like "Mark as not Ok", "send it back", "Check recommendation", "Mark as Ok", "Proceed with claim", "Receive request for new hardware", "Analyze request", "Approve request", "Deny request", "inform employee via email", "edit request", "Receive offer", "fill in form", "decide if approved", "this is accepted", "receive hardware by IT department", "take care of invoice" which are associated with the LTL constraints of Init, Response, and End depending on the specific order and conditions mentioned in each sentence.

Prompt:

do the same for the following:

71 In the scenario where the management does not approve the request, it can send the request back to the IT department for rework, then the IT department has to go back to editing and completing a request.

72 If the management does not approve the request, it can also decide to deny the request after which the process is ended.

73 The loan approval process starts by receiving a customer request for a loan amount.

74 If the customer needs further review or the loan amount is for \$10,000 or more, the request is sent to the approver Web service.

75 The process of an Office Supply Request starts when any employee of the organization submits an office supply request.

76 Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the office supplies.

77 If the request is rejected, the process will end.

78 If the request is asked to make a change, then it is returned to the petitioner who can review the comments for the change request.

79 If the request is approved, it will go to the purchase department that will be in charge of making quotations (using a subprocess) and select a vendor.

80 After a vendor is selected and confirmed, the system will generate and send a purchase order and wait for the product to be delivered and the invoice received.

Response:
Sure, let's analyze these sentences:

"In the scenario where the management does not approve the request, it can send the request back to the IT department for rework, then the IT department has to go back to editing and completing a request."

Activities:

approval, rejection, change: A

Send back request to IT: B

edit request: C

LTL constraint: Response (A, B), Response (B, C)

"If the management does not approve the request, it can also decide to deny the request after which the process is ended."

Activities:

approval, rejection, change: A

Deny request: B

LTL constraint: Response (A, B), End(B)

"The loan approval process starts by receiving a customer request for a loan amount."

Activities:

receive customer request: A

LTL constraint: Init (A)

"If the customer needs further review or the loan amount is for $10,000 or more, the request is sent to the approver Web service."

Activities:

need further review: A

submit office supply request: B

LTL constraint: Response (A, B)

"The process of an Office Supply Request starts when any employee of the organization submits an office supply request."

Activities:

submit office supply request: A

LTL constraint: Init (A)

"Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the office supplies."

Activities:

register requirement: A

receive request: B

LTL constraint: Response (A, B)

"If the request is rejected, the process will end."

Activities:

reject request: A

LTL constraint: End (A)

"If the request is asked to make a change, then it is returned to the petitioner who can review the comments for the change request."

Activities:

ask to make change: A

return to petitioner/employee: B

LTL constraint: Response (A, B)

"If the request is approved, it will go to the purchase department that will be in charge of making quotations (using a subprocess) and select a vendor."

Activities:

approval, rejection, change: A

go to purchase department: B

select vendor: C

LTL constraint: Response (A, B), Response (B, C)

"After a vendor is selected and confirmed, the system will generate and send a purchase order and wait for the product to be delivered and the invoice received."

Activities:

select vendor: A

generate purchase order: B

process delivery: C

receive invoice: D

LTL constraint: Response (A, B), Response (B, C), Response (C, D)

These sentences have activities like "approval, rejection, change", "Send back request to IT", "edit request", "Deny request", "receive customer request", "need further review", "submit office supply request", "register requirement", "receive request", "reject request", "ask to make change", "return to petitioner/employee", "go to purchase department", "select vendor", "generate purchase order", "process delivery", "receive invoice" which are associated with the LTL constraints of Init, Response, and End depending on the specific order

Prompt:
do the same for the following:

80 After a vendor is selected and confirmed, the system will generate and send a purchase order and wait for the product to be delivered and the invoice received.

81 In any of the cases, approval, rejection or change required the system will send the user a notification.

82 The process starts when any employee of the organization submits a vacation request.

83 Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the vacation.

84 If the request is rejected, the application is returned to the applicant/employee who can review the rejection reasons.

85 If the request is approved, a notification is generated to the Human Resources Representative, who must complete the respective management procedures.

86 The process starts by reviewing the order.

87 If something is missing, the staff has to inform customer about a potential delay and order the missing products.

88 After waiting an average of x days for the delivery, the staff should check again if all products are on store.

89 The courier company informs the client about a potential delay until there are enough human resources to process the delivery.

90 Once the driver delivers the package, she should update the system so all stakeholders are notified.

Response:
Sure, let's analyze these sentences:

"After a vendor is selected and confirmed, the system will generate and send a purchase order and wait for the product to be delivered and the invoice received."

Activities:

select vendor: A

generate purchase order: B

process delivery: C

receive invoice: D

LTL constraint: Response (A, B), Response (B, C), Response (C, D)

"In any of the cases, approval, rejection or change required the system will send the user a notification."

Activities:

approval, rejection, change: A

generate notification: B

LTL constraint: Response (A, B)

"The process starts when any employee of the organization submits a vacation request."

Activities:

submit vacation request: A

LTL constraint: Init (A)

"Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the vacation."

Activities:

register requirement: A

receive request: B

LTL constraint: Response (A, B)

"If the request is rejected, the application is returned to the applicant/employee who can review the rejection reasons."

Activities:

reject request: A

return application: B

LTL constraint: Response (A, B)

"If the request is approved, a notification is generated to the Human Resources Representative, who must complete the respective management procedures."

Activities:

approval, rejection, change: A

generate notification: B

manage case: C

LTL constraint: Response (A, B), Response (B, C)

"The process starts by reviewing the order."

Activities:

review order: A

LTL constraint: Init (A)

"If something is missing, the staff has to inform customer about a potential delay and order the missing products."

Activities:

need further review: A

inform about delay: B

order missing products: C

LTL constraint: Response (A, B), Response (B, C)

"After waiting an average of x days for the delivery, the staff should check again if all products are on store."

Activities:

wait for x days: A

check again: B

LTL constraint: Response (A, B)

"The courier company informs the client about a potential delay until there are enough human resources to process the delivery."

Activities:

inform about delay: A

process delivery: B

LTL constraint: Response (A, B)

"Once the driver delivers the package, she should update the system so all stakeholders are notified."

Activities:

deliver package: A

update system: B

LTL constraint: Response (A, B)

These sentences have activities like "select vendor", "generate purchase order", "process delivery", "receive invoice", "approval, rejection, change", "generate notification", "submit vacation request", "register requirement", "receive request", "reject request", "return application", "manage case", "review order", "need further review", "inform about delay", "order missing products", "wait for x days", "check again", "deliver package", "update system" which are associated with the LTL constraints of Init and Response depending on the specific order.

Prompt:

do the same for the following:

91 Whenever the sales department receives an order, a new process instance is created.

92 If the storehouse has successfully reserved or back-ordered every item of the part list and the preparation activity has finished, the engineering department assembles the bicycle.

93 Afterwards, the sales department ships the bicycle to the customer and finishes the process instance.

94 If the requested amount is lower or equal to 1M$, the company assess the risk of the credit.

95 After the approval request, the credit could be accepted or rejected

96 If the requested amount is greater than 1M$, an approval must be requested.

97 If the requested amount is lower or equal to 1M$, the company assess the risk of the credit.

98 As soon as an employee files an expense report, the process is started.

99 Once the expense report is received, a check is performed to make sure the employee has an account.

100 If this check does not take place within 28 days, the process is stopped and the employee is sent a submit email.

101 First, the supervisor approves within 7 days, this will result in the deposit of the reimbursement directly to the employee's bank account.

102 Second, the supervisor rejects within 7 days, as a result the employee receives a notification through email that the request has been rejected.

103 Third, if it takes the supervisor more than 7 days to respond, an email is sent to the employee informing him/her that the approval is in progress and the part of the processes after a supervisor is asked for approval is repeated.

Response:
Sure, let's analyze these sentences:

"Whenever the sales department receives an order, a new process instance is created."

Activities:

receive order: A

start process instance: B

LTL constraint: Response (A, B)

"If the storehouse has successfully reserved or back-ordered every item of the part list and the preparation activity has finished, the engineering department assembles the bicycle."

Activities:

reserve/back-order items: A

finish preparation: B

assemble bicycle: C

LTL constraint: Response (A, B), Response (B, C)

"Afterwards, the sales department ships the bicycle to the customer and finishes the process instance."

Activities:

ship bicycle: A

finish process instance: B

LTL constraint: Response (A, B)

"If the requested amount is lower or equal to 1M$, the company assess the risk of the credit."

Activities:

request amount <=1M: A

assess risk: B

LTL constraint: Response (A, B)

"After the approval request, the credit could be accepted or rejected"

Activities:

approval request: A

accept/reject credit: B

LTL constraint: Response (A, B)

"If the requested amount is greater than 1M$, an approval must be requested."

Activities:

request amount >1M: A

request approval: B

LTL constraint: Response (A, B)

"If the requested amount is lower or equal to 1M$, the company assess the risk of the credit."

Activities:

request amount <=1M: A

assess risk: B

LTL constraint: Response (A, B)

"As soon as an employee files an expense report, the process is started."

Activities:

file expense report: A

start process: B

LTL constraint: Response (A, B)

"Once the expense report is received, a check is performed to make sure the employee has an account."

Activities:

receive expense report: A

perform check: B

LTL constraint: Response (A, B)

"If this check does not take place within 28 days, the process is stopped and the employee is sent a submit email."

Activities:

check account status: A

stop process: B

send submit email: C

LTL constraint: Response (¬A, B), Response (B, C), Within (¬A, B, 28)

"First, the supervisor approves within 7 days, this will result in the deposit of the reimbursement directly to the employee's bank account."

Activities:

supervisor approval: A

deposit reimbursement: B

LTL constraint: Response (A, B), Within (A, B, 7)

"Second, the supervisor rejects within 7 days, as a result the employee receives a notification through email that the request has been rejected."

Activities:

supervisor rejection: A

send rejection notification: B

LTL constraint: Response (A, B), Within (A, B, 7)

"Third, if it takes the supervisor more than 7 days to respond, an email is sent to the employee informing him/her that the approval is in progress and the part of the processes after a supervisor is asked for approval is repeated."

Activities:

supervisor response >7 days: A

send email about progress: B

LTL constraint: Response (A, B), After (A, B, 7)

These sentences have activities like "receive order", "start process instance", "reserve/back-order items", "finish preparation", "assemble bicycle", "ship bicycle", "finish process instance", "request amount <=1M", "assess risk", "approval request", "accept/reject credit", "request amount >1M", "request approval", "file expense report", "start process", "receive expense report", "perform check", "check account status", "stop process", "send submit email", "supervisor approval", "deposit reimbursement", "supervisor rejection",

"send rejection notification", "supervisor response >7 days", "send email about progress" which are associated with the LTL constraints of Response, Within, and After depending on the specific order.